

SMART INDIA HACKATHON 2026

Bhoomi Setu

Real-Time National Land Acquisition & Management System

OFFICIAL IDEA SUBMISSION PARTICULARS

- Problem Statement ID: **SIH1608 / PS-GOV-2026**
- Problem Statement Title: **Real-Time National Land Acquisition & Management System**
- Theme: **Smart Governance & Infrastructure Automation (PM Gati Shakti Alignment)**
- PS Category: **Software (Web, Mobile Responsive & Cadastral GIS Platform)**
- Team Name: **Registered Team Name on Portal**
- Core Platform: **Bhoomi Setu - Digital 9-Stage Statutory Land Acquisition & DBT Ledger**
- Live Interactive Prototype: **<https://bruhatipuni.github.io/Real-Time-National-Land-Acquisition-Management-System/>**

01. ULPIN Map
02. Khata Check
03. Proposal
04. Sec 3A Notice
05. Sec 3C Dispute
06. Sec 3D Vesting
07. Sec 3G Award
08. DBT Handover
09. Corridor

1. DETAILED PROPOSED SOLUTION

- End-to-End 9-Stage Lifecycle: Digitizes land identification, khata verification, gazette notices (3A/3D), 100% solatium award (3G), and DBT handover (3E).
- Spatial Bhu-Aadhaar (ULPIN): Binds each parcel with its 14-digit national ULPIN, Survey of India geo-coordinates, and high-res drone orthomosaic video logs.
- Statutory RFCTLARR 2013 Engine: Automated calculation engine enforces base circle rate, 1.5x rural multiplier, compulsory 100% solatium, and 12% annual interest.
- Direct Benefit Transfer (DBT): Direct integration with PFMS / RBI e-Kuber for electronic fund credit directly into Aadhaar-linked beneficiary bank accounts.

2. HOW IT ADDRESSES THE PROBLEM

- Eliminates Bureaucratic Delays: Real-time SLA countdown timers track every statutory milestone, preventing file stagnation in revenue tehsils.
- 100% Landowner Transparency: Citizens view their exact parcel boundary on satellite maps, itemized compensation math, and bank credit receipts.
- Reduces Court Stay Orders: Interactive Section 3C dispute module allows landowners to file objections online, resolving disputes before court escalation.
- PM Gati Shakti Alignment: Unified corridor portfolio dashboard for Central Ministries across Highway, Dedicated Freight, and Energy corridors.

3. INNOVATION & NOVELTY

- Context-Aware AI Assistant (Bhu-Sathi): Syncs directly with the citizen's active parcel to answer specific questions regarding compensation, hearing dates, and next steps.
- Decision Support 'What Happens Next?': Dynamic banner on every page that provides immediate next action, responsible authority, and estimated timeframe.
- Apply for Land Papers Online: Citizens can apply for certified Jamabandi RoR, ULPIN maps, and Non-Encumbrance certificates with instant digital verification.
- Multi-Role RBAC Portals: 5 tailored interfaces for Ministry, LAO Collector, Field Surveyor, Citizen, and Implementing Agencies.

TECHNOLOGIES & FRAMEWORKS

- Frontend UI**
React 19, Vite, Tailwind CSS, Framer Motion, Recharts
- GIS Spatial Engine**
Leaflet, React-Leaflet, OpenStreetMap, Esri Satellite Imagery, Drone Payloads
- Backend REST API**
Node.js (v20+), Express.js, RESTful Architecture, Mongoose (MongoDB)
- Security & RBAC**
JWT Token Authentication, Role-Based Access Control (5 Personas), SHA-256 Hashes
- Client Services**
jsPDF (Client-Side PDF Generation), HTML5 Web Speech API (Read Aloud)
- Accessibility**
Bilingual (English / Hindi), Text Scaling (A-/A+), High-Contrast Mode

SYSTEM ARCHITECTURE & METHODOLOGY

Tier 1: Client Experience Layer

Role-tailored interfaces for Central Ministry, LAO Collector, Field Surveyor, Landowner Citizen, and Implementing Agencies with responsive GIS map and speech synthesis.



Tier 2: Business & Statutory Engine

Core workflow state machine enforcing RFCTLARR 2013 rules, statutory objection period validation (21-day window), solatium computation, and AI dispute radar.



Tier 3: Spatial & Data Layer

14-digit ULPIN Bhu-Aadhaar coordinates, cadastral polygons, orthomosaic drone video feeds, and tamper-evident audit logs with cryptographic hash verification.



Tier 4: National Integration Gateway

Pre-configured endpoints connecting with PFMS / RBI e-Kuber for direct DBT payouts, e-Gazette notification services, and state land modern records (DILRMP).

FEASIBILITY ASSESSMENT SUMMARY:

100% technically feasible using proven web technologies (React 19 + Node.js) and legally viable under RFCTLARR 2013 and National Highways Act 1956.

POTENTIAL CHALLENGE & RISK	IMPACT ON PROJECT	BHOOMI SETU MITIGATION STRATEGY
Rural Digital Literacy Barrier	Landowners unable to navigate complex forms or read English statutory orders.	Bilingual toggle (English/Hindi), A+/A- font scaling, High Contrast mode, and Web Speech API 'Read Page Aloud' feature.
Paper Record & Cadastral Discrepancies	Area / spelling mismatch between Jamabandi and physical field boundary.	Survey of India orthomosaic drone video overlay and Field Surveyor Workbench resolving boundary discrepancies prior to Sec 3A.
Sensitive Financial & Identity Privacy	Exposure of Aadhaar, bank accounts, or landowner personal compensation records.	DPDPA 2023 Privacy Center enforcing strict RBAC: bank details accessible only to Treasury/PFMS; zero public disclosure.
Dispute Litigation & Court Stays	High Court petitions halting critical infrastructure corridor development.	Online Section 3C grievance filing with 72-hour turnaround, automated CALA hearing schedules, and pre-litigation settlement.

BEFORE VS. AFTER TRANSFORMATION MATRIX

Acquisition Timeline Before: 24 – 36 Months After: 6 – 9 Months	70% Faster
Compensation Calculation Before: Opaque manual registers After: Automated LARR 2013 math	100% Transparent
DBT Payment Disbursement Before: Cheques; 4-6 months wait After: Direct PFMS in 48-72 hrs	Zero Leakage
Grievance Redressal Before: 90-180 Days unacknowledged After: 72-Hour Guaranteed SLA	Full Accountability
Litigation & Stay Orders Before: 35% – 45% corridors delayed After: < 10% litigation dispute	75% Drop in Stays
Citizen Tehsil Commute Before: Multiple physical visits After: 24/7 Digital Citizen Portal	Zero Paperwork

QUANTITATIVE NATIONAL IMPACT

TIMELINE REDUCTION (MONTHS)



INR 10,000+ CRORES

Estimated national cost savings by preventing project delays and cost overruns on PM Gati Shakti corridors.

100% DBT VIA PFMS

Zero middleman commissions; statutory 100% Solatium credited directly to landowner bank accounts.

BROADER IMPACT DIMENSIONS:

- Social: Dignified compensation, R&R tracking, trust in govt.
- Legal: Pre-litigation hearing settlement within 60 days.
- Environmental: GIS green cover & eco-sensitive zone avoidance.
- National: Accelerated highway, rail, and renewable corridors.

STATUTORY ACTS & REGULATORY STANDARDS

1. RFCTLARR Act, 2013 (Land Acquisition Act)

Sections 26, 30, First & Second Schedules: Mandates base market valuation, 1.25x-2.0x rural multiplier factor, compulsory 100% solatium allowance, and 12% per annum interest from Sec 3A notification date.

2. The National Highways Act, 1956

Sections 3A (Intention Notice), 3C (Objection Hearings by CALA), 3D (Declaration of Vesting), 3E (Possession Notice), and 3G (Determination of Amount by Competent Authority).

3. Digital India Land Records Modernization (DILRMP)

Technical standard specifications for Unique Land Parcel Identification Number (ULPIN / Bhu-Aadhaar) 14-digit alphanumeric geocoding, Department of Land Resources, Ministry of Rural Development.

4. Public Financial Management System (PFMS)

Centralized DBT rules for electronic beneficiary account verification, Treasury disbursement mandates, and RBI e-Kuber clearing protocols.

5. Digital Personal Data Protection Act, 2023 (DPDPA)

Regulatory standards for citizen consent minimization, biometric tokenization, role-based data encryption, and tamper-evident audit logs.

LIVE PROTOTYPE & OPEN-SOURCE VERIFICATION

ONLINE DEPLOYED SYSTEM (GITHUB PAGES)

<https://bruhatipuni.github.io/Real-Time-National-Land-Acquisition-Management-System/>

Accessible 24/7 on all devices with one-click role logins for Super Admin, District Collector / LAO, Surveyor, and Citizen.

SOURCE CODE REPOSITORY (GITHUB)

<https://github.com/Bruhatipuni/Real-Time-National-Land-Acquisition-Management-System>

Active Branches: main (core production), citizen/local (15 citizen modules), surveyor (field GIS tools), gh-pages (live build).

SIH 2026 EVALUATION COMPLIANCE:

- Strictly adheres to the mandatory 6-slide presentation limit.
- Comprehensive bullet points, structured comparison matrices & graphs.
- Includes end-to-end working software prototype and live URL.
- Directly answers national priority: PM Gati Shakti & DILRMP.
- Ready to upload as PDF on the SIH Idea Submission portal.